Smart Waste Sorter System: Dataset Documentation Report

1. Dataset Overview

Purpose. The dataset teaches an AI model to recognise what material a piece of waste is made of, so that the Smart Waste Sorter can recommend the correct bin.

How it supports classification. The project uses a YOLO11 *image classification* model (`yolo11n-cls`). Classification means the model looks at one picture and gives one label, for example "Plastic", together with a confidence score. The dataset provides the labelled example pictures the model learns from. It is **not** an object-detection dataset, so there are no bounding boxes.

Waste types included. Plastic, Paper (including cardboard), Can / Metal, Glass and Organic waste.

Use in the system. The dataset is used only during training (`scripts/train_model.py`). The resulting model is saved as `model/best.pt`. When the application runs, the webcam image is classified by this model, and the result is converted into a bin recommendation, a voice announcement and a history record. The dataset images themselves are not used while the application is running.

2. Image Sources

Part of dataset	Source	How obtained	License / restrictions	Status
Plastic, Glass, Can_Metal, Paper (1,893 images)	TrashNet, `data/dataset-resized.zip`, repository `garythung/trashnet` (https://github.com/garythung/trashnet)	Downloaded from another dataset and copied with the prefix `trashnet_`	MIT License (Copyright 2017 Gary Thung), as recorded in `docs/dataset_sources.md`. The repository asks users to cite it.	Documented; the project's own record agrees with the file counts
Organic (5 images)	**Not provided**	Probably collected manually (file names such as `organic waste1.jfif`)	**Needs verification**	Source unknown

Part of dataset	Source	How obtained	License / restrictions	Status
Organic (about 411 images, "Food Organics")	RealWaste, UCI repository (https://archive.ics.uci.edu/dataset/908/realwaste)	Recorded in `docs/dataset_sources.md` as added on 2026-10-06	UCI page states CC BY 4.0, the authors' IEEE DataPort page states CC BY-NC-SA 4.0. The project follows the stricter one (non-commercial, share-alike).	**These images are not present in the project files.** The record and the files disagree.
Folder `dataset/DATASET/` (two classes `O` and `R`, 19,389 images)	**Not documented**	Unknown	**Needs verification**	Not used for training

Notes

1. The URLs above are copied from the project's own documentation (`docs/dataset_sources.md`). They were not opened or independently verified for this report.
2. TrashNet also contains a `trash` class. The project did not use it.
3. The Paper category was built from 299 TrashNet *paper* images and 201 TrashNet *cardboard* images, randomly sampled to 500 images (seed `20261006`).
4. The file `download_dataset.py` points to a Hugging Face dataset (`bryandts/waste_organic_anorganic_classification`). The log `error.txt` shows that this download **failed**, so no data from it is confirmed to be in the project.
5. In the few images opened for this report, the `O` folder showed fruit and nuts and the `R` folder showed plastic bottles. The meaning of the letters `O` and `R` is **not documented**.

3. Number of Images

3.1 Images in the training dataset (Verified)

Category	Training	Validation	Testing	Total
Can_Metal	286	62	62	410
Glass	351	75	75	501
Organic	3	1	1	5
Paper	350	75	75	500

Category	Training	Validation	Testing	Total
Plastic	338	72	72	482
Total	**1,328**	**285**	**285**	**1,898**

3.2 Other image folders (Verified)

Folder	Images	Role
`raw_images/`	1,898	Original copies of the same pictures. They are the source for `dataset/`, not extra data. `docs/dataset_manifest.csv` links each file (1,898 rows).
`dataset/DATASET/TRAIN/O`	12,565	Unused two-class folder
`dataset/DATASET/TRAIN/R`	6,824	Unused two-class folder
`dataset/DATASET/TEST/O`	0	Folder exists but is empty
`dataset/DATASET/TEST/R`	Folder missing	The `DATASET` folder appears incomplete (reason unknown)

3.3 Quality checks

Check	Result	Status
Corrupted or unreadable images	0 found in all dataset images (every file was opened)	Verified
Image format and size	`dataset/train	val
Duplicates inside a category	Removed by `prepare_dataset.py` (exact-copy check)	Script behaviour; not re-run
Identical images with different labels	**3 pairs found** (see below)	Verified
Missing files	None found. The manifest (1,898 rows) matches the folders.	Verified

The 3 conflicting pairs (same picture content, different labels, so the model receives contradictory answers):

Pair	Original TrashNet files	Location
1	`glass176` and `plastic152`	both in `test/`
2	`glass115` and `metal91`	`train/Glass` and `val/Can_Metal`
3	`plastic332` and `glass389`	`train/Plastic` and `val/Glass`

Pairs 2 and 3 also place the same picture in both the training and validation sets, which slightly inflates validation accuracy. Each pair should be inspected by eye and one copy removed or relabelled.

4. Waste Categories

Only categories that exist in the dataset folders are listed.

Dataset label	Meaning and typical objects	System display name	Notes
`Plastic`	Plastic bottles, containers and packaging (TrashNet `plastic`)	Plastic	482 images
`Paper`	Paper and cardboard (TrashNet `paper` and `cardboard` merged)	Paper	500 images. The project also has a `Cardboard` entry in `config.py`, but the model has no such class.
`Can_Metal`	Metal items such as drink cans (TrashNet `metal`)	Metal can	410 images. TrashNet's `metal` class may include metal objects other than cans (**needs verification**).
`Glass`	Glass bottles and jars (TrashNet `glass`)	Glass	501 images
`Organic`	Food and compostable waste	Organic waste	**Only 5 images.** Too few to train reliably.

Label consistency check. The five dataset folder names, the five model classes (read from `model/best.pt`) and the `CATEGORIES` list in `src/config.py` are identical. No name mismatch was found.

Category expected but missing: `Other`. The scripts `train_model.py` and `prepare_dataset.py` and the README expect a sixth class, `Other` (non-waste objects, people and empty scenes). No `Other` folders exist, and the current model was trained without it. `Other` is therefore **not** a category of the current dataset.

5. Dataset Organization

The project uses **classification folders**, not YOLO bounding-box files. Each class has its own folder inside each split, which is the layout Ultralytics expects for classification.

```
smart_waste/
├── raw_images/          original images, one folder per category
|   ├── Can_Metal/  Glass/  Organic/  Paper/  Plastic/
├── dataset/
|   ├── train/           images used to train the model
|   |   └── Can_Metal/  Glass/  Organic/  Paper/  Plastic/
|   ├── val/             images used to check the model during training
|   |   └── (same five folders)
|   ├── test/            images kept for a final check
|   |   └── (same five folders)
|   ├── DATASET/         unused two-class folder (TRAIN/O, TRAIN/R, empty
TEST/O)
|   ├── train.cache      automatic speed-up file made by Ultralytics
|   └── val.cache        automatic speed-up file made by Ultralytics
├── docs/
|   ├── dataset_manifest.csv   record of every image: new name, category,
split, original name
|   └── dataset_sources.md     sources and licenses
├── scripts/prepare_dataset.py  builds the splits from raw_images/
├── scripts/train_model.py      trains the model
└── model/best.pt               the trained classifier
```

Item	Purpose
`raw_images/`	Keeps the original pictures so the dataset can be rebuilt.

Item	Purpose
`dataset/train`, `val`, `test`	The three splits. Each contains one folder per class; the folder name is the label.
`*.cache` files	Created automatically by Ultralytics. They can be deleted safely.
`docs/dataset_manifest.csv`	Traceability: shows which original image became which file in which split.
`data.yaml`	**Not present and not needed.** Classification training uses `data=dataset/` (confirmed in `runs/.../args.yaml`).
Label (`.txt`) files	**None.** Classification does not use them.

File names follow the pattern `<category>_<number>.jpg` (for example `glass_0044.jpg`). `prepare_dataset.py` converts every image to JPEG.

6. Training, Validation and Testing Splits

Definitions

- **Training set:** the pictures the model learns from.
- **Validation set:** pictures used during training to check progress and choose the best saved model. The model does not learn from them directly.
- **Testing set:** pictures kept aside for a final, unbiased check of the finished model.

Actual split (Verified)

Split	Images	Percentage of 1,898
Training	1,328	70.0%
Validation	285	15.0%
Testing	285	15.0%
Total	**1,898**	**100%**

The split is therefore about **70 / 15 / 15**. `prepare_dataset.py` sets `VAL_RATIO = 0.15` and `TEST_RATIO = 0.15` with a fixed random seed of 42, which makes the split repeatable. Images are grouped so that photos of the same item stay in one split.

Organic is the exception. Its 5 images are split 3 / 1 / 1 (60% / 20% / 20%). One validation image and one test image cannot measure anything meaningful.

Recorded results (not a final evaluation). The training run `waste_sorter_20261008_132810` (30 epochs) recorded a best validation top-1 accuracy of **95.4% (272 of 285 images)**, and 94.4% at the last epoch. These figures come from the validation set only. The test set has **not** been used in any recorded evaluation. Because the validation set contains one Organic image, these results say almost nothing about Organic waste.

Note. Object-detection datasets would need one annotation file per image. This project does not, because it uses classification.

7. Dataset Preparation Process

Step	Task	Status	Evidence
1	Collect or download images	**Partly complete.** TrashNet images done. Organic images mostly missing.	`raw_images/`, `dataset_sources.md`
2	Check quality and remove unusable files	**Complete (with one gap).** The script removes exact duplicates and broken files within each category. It does not detect the same picture under two different labels.	`prepare_dataset.py`; verified checks in section 3.3
3	Organise images by category	**Complete**	`raw_images/` and `dataset/` folders

Step	Task	Status	Evidence
4	Annotate images	**Not required** (classification uses folder names)	Folder layout
5	Divide into training, validation and testing sets	**Complete** (about 70/15/15)	`dataset_manifest.csv`, folder counts
6	Check folder paths and configuration	**Partly complete.** Folder layout is valid for the four main classes. Training fails the script's own check because the `Other` folders are missing. No `data.yaml` is needed.	`train_model.py` (`validate_dataset`)
7	Verify the dataset is ready for training	**Pending** (not ready; see section 9)	This report

Building steps already carried out: images were copied from the sources, converted to JPEG at quality 95, renamed, split and recorded in the manifest.

Pending tasks

1. Add enough Organic images (the script suggests at least 50 training and 10 validation and test images per class).
2. Decide about the `Other` class and either add its images or remove it from the scripts and README.
3. Fix the three conflicting image pairs.
4. Confirm the source and license of the Organic images and of the `DATASET/` folder.
5. Re-run `prepare_dataset.py`, then re-train and evaluate on the test set.

8. Dataset Summary

Item	Details
Dataset name	No single name. TrashNet (resized) plus 5 Organic images.
Image sources	TrashNet (garythung/trashnet, MIT License). Organic images: source not provided, needs verification. `DATASET/` O/R folder: source not documented.
Total number of images	1,898 in use (verified). The `raw_images/` folder holds the same 1,898 as copies. A further 19,389 images in `DATASET/` are unused.
Waste categories	Can_Metal, Glass, Organic, Paper, Plastic. (`Other` is expected by the scripts but does not exist.)
Images per category	Can_Metal 410, Glass 501, Organic 5, Paper 500, Plastic 482
Dataset format	Image classification, JPEG, one folder per class inside train / val / test
Annotation format	Folder names are the labels. No YOLO label files.
Training split	1,328 images (70.0%)
Validation split	285 images (15.0%)
Testing split	285 images (15.0%). Not yet used in any recorded evaluation.
Dataset configuration file	None (`data.yaml` is not required). Training used `data=dataset/`.
Current preparation status	Ready for four classes. **Not ready** for the full five-class goal, because Organic has only 5 images. Retraining with the current scripts is blocked until the `Other` class question is resolved.

9. Conclusion

The dataset is the foundation of the Smart Waste Sorter. The model can only recognise materials that appear in the training pictures, so the quality and balance of the data directly control how reliable the sorting will be.

Readiness. For Plastic, Paper, Can_Metal and Glass, the dataset is well organised: each class has 410 to 501 images, the split is about 70/15/15, there are no corrupted files, and every image is traceable through the manifest. For Organic waste, the dataset is **not ready**, because it has only 5 images. The current model's accuracy figure of 95.4% should not be quoted as evidence that all five categories work.

Limitations and missing information

- Organic images: source and license not provided; the images documented as added (RealWaste, about 411) are not in the project.
- Source of the `DATASET/` folder is undocumented, and its test part is empty.
- Three images carry conflicting labels.
- The `Other` class is expected by the scripts but missing from the data.
- TrashNet images are close-up photos taken in similar conditions. Performance on real webcam images has not been measured (**needs verification**).
- No test-set evaluation has been recorded.

Recommended next steps

1. Add at least 50 varied Organic training images (and 10 each for validation and testing), with a documented source and license.
2. Take extra photos with the actual webcam and bin setup for all classes.
3. Resolve the `Other` class (add images or remove it from the code and README).
4. Correct the three conflicting image pairs.
5. Retrain, then evaluate on the **test** split and record per-class results.
6. Update `docs/dataset_sources.md` so that it matches the files actually present.